

Quiz 1

Problem Solving, Math 1000 – Summer 2016

Thursday, June 2

The following quiz is out of 40 points, and will count towards your quiz grade. Each problem is worth exactly 10 points. Answer the following problems (you may use your notes).

Problem 1 *Draw a network which has exactly three spanning trees.*

Problem 2 *List some property/properties of a graph which are sufficient to categorize it as a network. List some property/properties of a graph which are sufficient to categorize it as a tree. Draw an example of each.*

Problem 3 *Explain how you would go about finding a MaxST in a given weighted network. If there is a theorem that relates MSTs to MaxSTs, it is sufficient for you to state that and explain which algorithm you could then use.*

Problem 4 *Give a real-world example of a weighted network and a real-world interpretation of a MST in that network.*